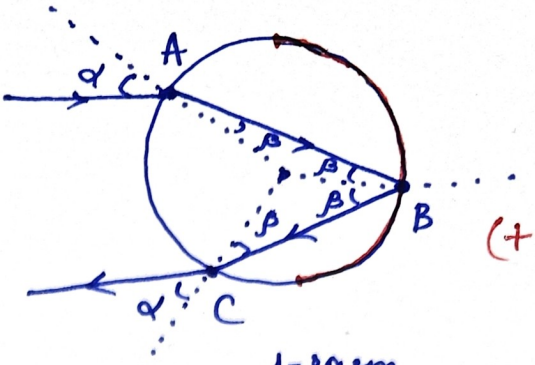
5- masala

Nurning shisha shar ichidagi yoli 1-rasmda ko'rsatilgan.



1-rasm

(+2 ball)

a) Nurning dastlabki yo'nalishdan og'ishi δ, uning A, B va C nuqtalardagi burilishlar yig'indisiga teng

$$\delta = (\alpha-\beta) + (\pi - 2\beta) + (\alpha-\beta)$$

$$= \pi + 2\alpha - 4\beta \quad (+3 \text{ ball})$$

tushish burchagining kichik qiymatlarida

$$\frac{\sin\alpha}{\sin\beta} = n \Rightarrow \frac{\alpha}{\beta} = n, \text{ bundan } \beta = \frac{\alpha}{n} \quad (+2 \text{ ball})$$

Burilish burchagi:

$$\delta = \pi + 2\alpha - \frac{4\alpha}{n} = \pi + \left(2 - \frac{4}{n}\right)\alpha \quad (+1 \text{ ball})$$

b). Shishaga tushayotgan parallel nurlar dastasi shishani tark etganida yana parallel nurlar dastasi bo'lishi uchun barcha nurlar uchun δ bir xil bo'lishi kerak. δ – tushish burchagiga bog'liq bo'lmasligi kerak

$$\delta = \pi + \underbrace{\left(2 - \frac{4}{n}\right)}_{=0} \cdot \alpha$$

$2 - \frac{4}{n} = 0$, bundan $n = 2$ ekanini topamiz. (+2 ball)